Power Budget Example

Team Number:	208
Project Name:	**Self-Balancing Work Bench**
Team Member Names:	**Joseph, Caden, Emma**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	IMU	ICM-42688-P	+1.71V - +30V	1		0.88	mA
	Voltage Level Shifter	SN74LVC125A	1.65V - 3.6V	1		2.02	mA
	Nano Curiousity	PIC18F57Q43	+1.8V - +5V	1		12	mA
						14.9	mA
c2. Regulator or Source Choice	+5V Regulator	LM7805	+7V - +25V	1	1000	1000	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.

+5V Power Rail	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	Curiosity Nano	PIC18F57Q43	+1.8V - +5V	1		12	mA
					Subtotal	12	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	15	mA
					Total Remaining Current Available on 5V Rail	985	mA
c4. Regulator or Source Choice	+3.3V Voltage Regulator	LP2950	+2V - +30V	1	100	100	mA
+3.3V Power Rail	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	IMU	ICM-42688-P	1.71V - 3.6V	1		0.88	mA
	Voltage Level Shifter	SN74LVC125A	1.65V - 3.6V	1		2.02	mA
						0	mA
						0	mA
					Subtotal	2.9	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	3.625	mA
					Total Remaining Current Available on 3.3V Rail	96.375	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails

External Power Source 1	**Component Name**	**Part Number**	**Supply**	**Output Voltage**	**Absolute**	**Total**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	YNQX09L090065UL	110VAC	+9V	650	650	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	+7V - +25V	+5V	1000	22.9	mA
					Total Remaining Current Available on External Power Source 1	627.1	mA

Voltage:	3.3	mA
	ICM-42688-P	0.88
	SN74LVC125A	2.02
	Total (mA):	2.9
	Total (mW):	9.57

Voltage:	1.7	mA
	LP2950	2.90
	Total (mA):	2.90
	Total (mW):	4.93

Voltage Regulator Power Dissipations

Voltage:	5	mA
Supplies (mW):	14.50	2.90
	PIC18F57Q43	12.00
	Total (mA):	14.90
	Total (mW):	74.50

Voltage:	4	mA
	LM7805	14.90
	Total (mA):	14.90
	Total (mW):	59.60

Voltage Regulator Power Dissipation

Voltage:	9	mA
Supplies (mW)	134.10	14.90
Bias Supply	LM7805	8
	Total (mA):	22.90
	Total (mW):	206.10